# Thibault de Surrel

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## Education

LAMSADE, Univ. Paris Dauphine - PSL (Paris, France), PhD in Computer Science 2023 – 2026

- Title of the PhD: Wrapping Around the Manifold: Geometric Statistics for Brain-Computer Interfaces.
- Supervisors: Florian Yger, Sylvain Chevallier and Fabien Lotte.

ENS Paris-Saclay (Gif-sur-Yvette, France), Master 2 MVA: Mathematics, Vision, Learning 2022 – 2023

- Master focused on the mathematical aspects of data science, numerical data acquisition, processing and automatic interpretation.

ENS Paris-Saclay (Gif-sur-Yvette, France), Master 2 Formation pour l'enseignement supérieur 2021 – 2022

- Master of Science in Mathematics teaching.
- Agrégation externe de Mathématiques.

ENSTA Paris (Palaiseau, France), Engineer's Degree - Diplôme d'Ingenieur 2019 – 2023

- ENSTA Paris is amongst the top 5 engineering schools in France. Major in applied mathematics.

Lycée Claude Fauriel (St-Etienne, France), Preparatory class - Classes préparatoires 2017 – 2019

- Intensive two-year course in mathematics and physics to prepare for National exams to enter top French engineering schools.

## Internships

Research internship, LAMSADE, Univ. Paris Dauphine - PSL (Paris, France) Apr. 2023 – Sep. 2023

Supervised by: Florian Yger, Sylvain Chevallier and Fabien Lotte

- Matching trajectories of covariance matrices on the Riemannian manifold of symmetric definite positive (SPD) matrices with application on EEG data.
- Building Riemannian versions of the t-SNE and MDS algorithms to visualize high-dimensional SPD matrices.

Research internship, DataShape team, Inria (Sophia-Antipolis, France) May 2021 – Aug. 2021

Supervised by: Mathieu Carrière

- Build a neural network to predict persistence images given a point cloud.
- Build a VAE to learn the joint distribution of persistence images and point clouds.

## Publications

A probabilistic view on Riemannian machine learning models for SPD matrices Oct. 2025

Thibault de Surrel, Florian Yger, Fabien Lotte, and Sylvain Chevallier

Geometric Science of Information (GSI 2025)

Interpretability of Riemannian tools used in brain computer interfaces Aug. 2025

Thibault de Surrel, Tristan Venot, Marie-Constance Corsi, and Florian Yger

IEEE 35th International Workshop on Machine Learning for Signal Processing (MLSP 2025)

Wrapped Gaussian on the manifold of Symmetric Positive Definite Matrices Jul. 2025

Thibault de Surrel, Fabien Lotte, Sylvain Chevallier, and Florian Yger

ICML 2025

Geometry-Aware visualization of high dimensional SPD matrices Fev. 2025

Thibault de Surrel, Sylvain Chevallier, Fabien Lotte and Florian Yger

Transactions on Machine Learning Research (TMLR)

Averaging trajectories on the manifold of symmetric positive definite matrices — Aug. 2024

Thibault de Surrel, Florian Yger, Sylvain Chevallier and Fabien Lotte

EUSIPCO 2024

RipsNet: a general architecture for fast and robust estimation of the persistent homology of point clouds — Apr. 2022

Thibault de Surrel, Felix Hensel, Mathieu Carrière, Théo Lacombe, Yuichi Ike, Hiroaki Kurihara, Marc Glisse and Frédéric Chazal

ICLR 2022 Workshop on Geometrical and Topological Representation Learning

## Teaching

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|---|---|
| Linear Model and its Generalization, First year Master, Univ. Paris Dauphine - PSL | 2025 |
| Fundamental of ML, Third year Bachelor, Univ. Paris Dauphine - PSL | 2024 |
| Mathematical statistics, Third year Bachelor, Univ. Paris Dauphine - PSL | 2024-2025 |
| Algebra and numerical methods, Second year Bachelor, Univ. Paris Dauphine - PSL | 2024-2025 |